

# Summarizing Data

## Computational Mathematics and Statistics

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# One Minute Paper Results

What was the most important thing you learned during this class?



What important question remains unanswered for you?



# Data Types / Descriptives / Visualizations

| Data Type                  | Descriptive Stats                             | Visualization                |
|----------------------------|-----------------------------------------------|------------------------------|
| Continuous                 | mean, median, mode, standard deviation, IQR   | histogram, density, box plot |
| Discrete                   | contingency table, proportional table, median | bar plot                     |
| Categorical                | contingency table, proportional table         | bar plot                     |
| Ordinal                    | contingency table, proportional table, median | bar plot                     |
| Two quantitative           | correlation                                   | scatter plot                 |
| Two qualitative            | contingency table, chi-squared                | mosaic plot, bar plot        |
| Quantitative & Qualitative | grouped summaries, ANOVA, t-test              | box plot                     |

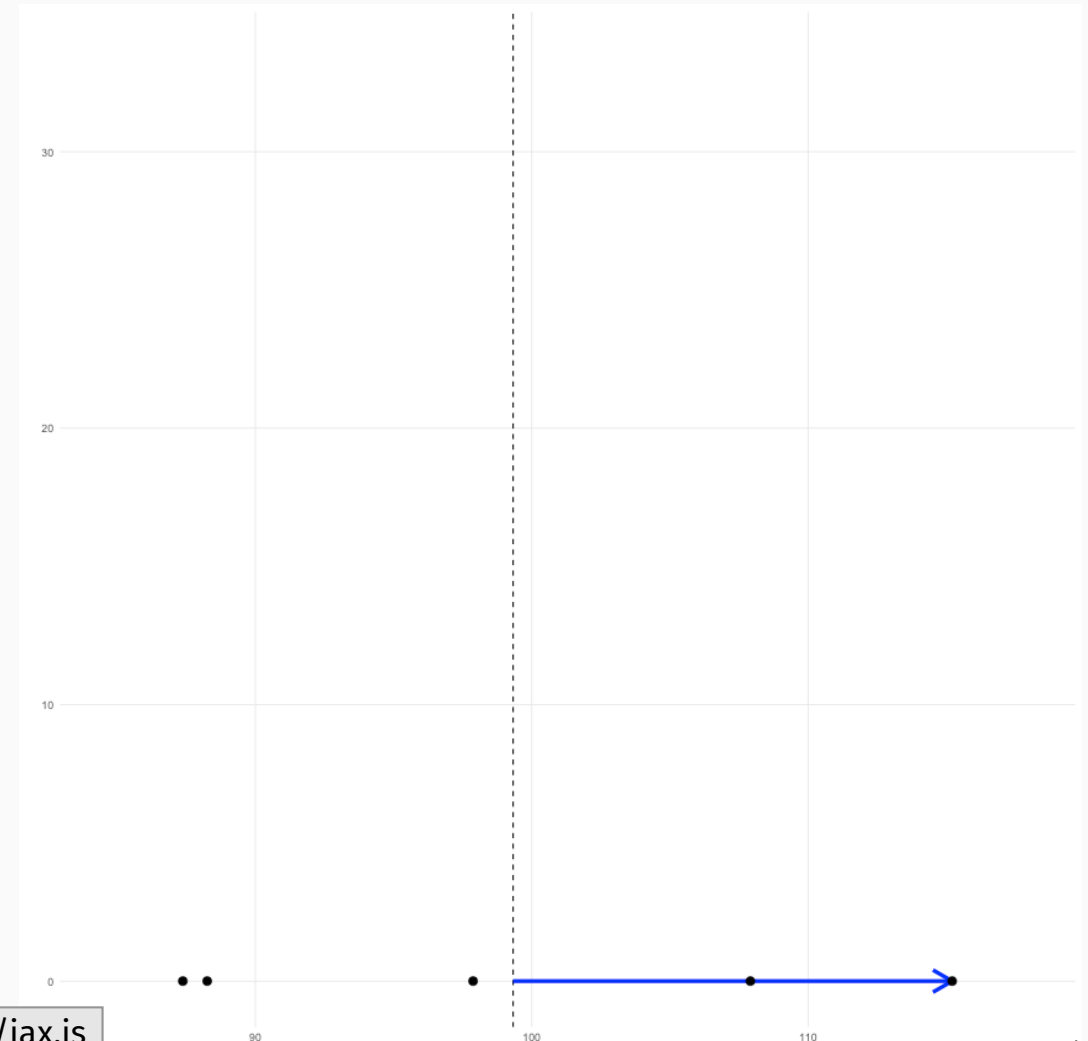
# Variance

Population Variance: *[Math Processing Error]*  
Consider a dataset with five values (black points in the figure). For the largest value, the deviance is represented by the blue line (*[Math Processing Error]* ).

See also:

<https://shiny.rit.albany.edu/stat/visualizess/>

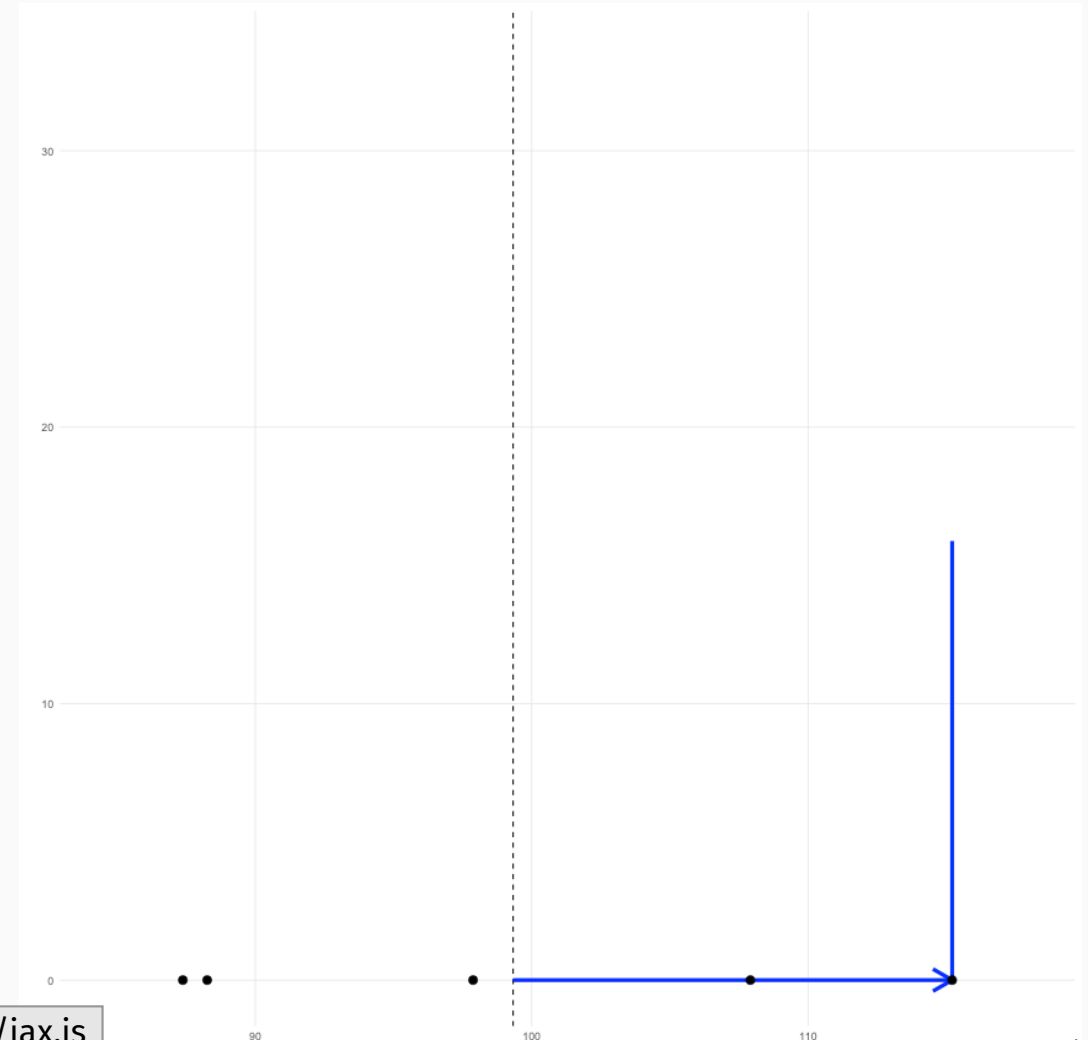
<https://github.com/jbryer/VisualStats/>



# Variance (cont.)

Population Variance: *[Math Processing Error]*

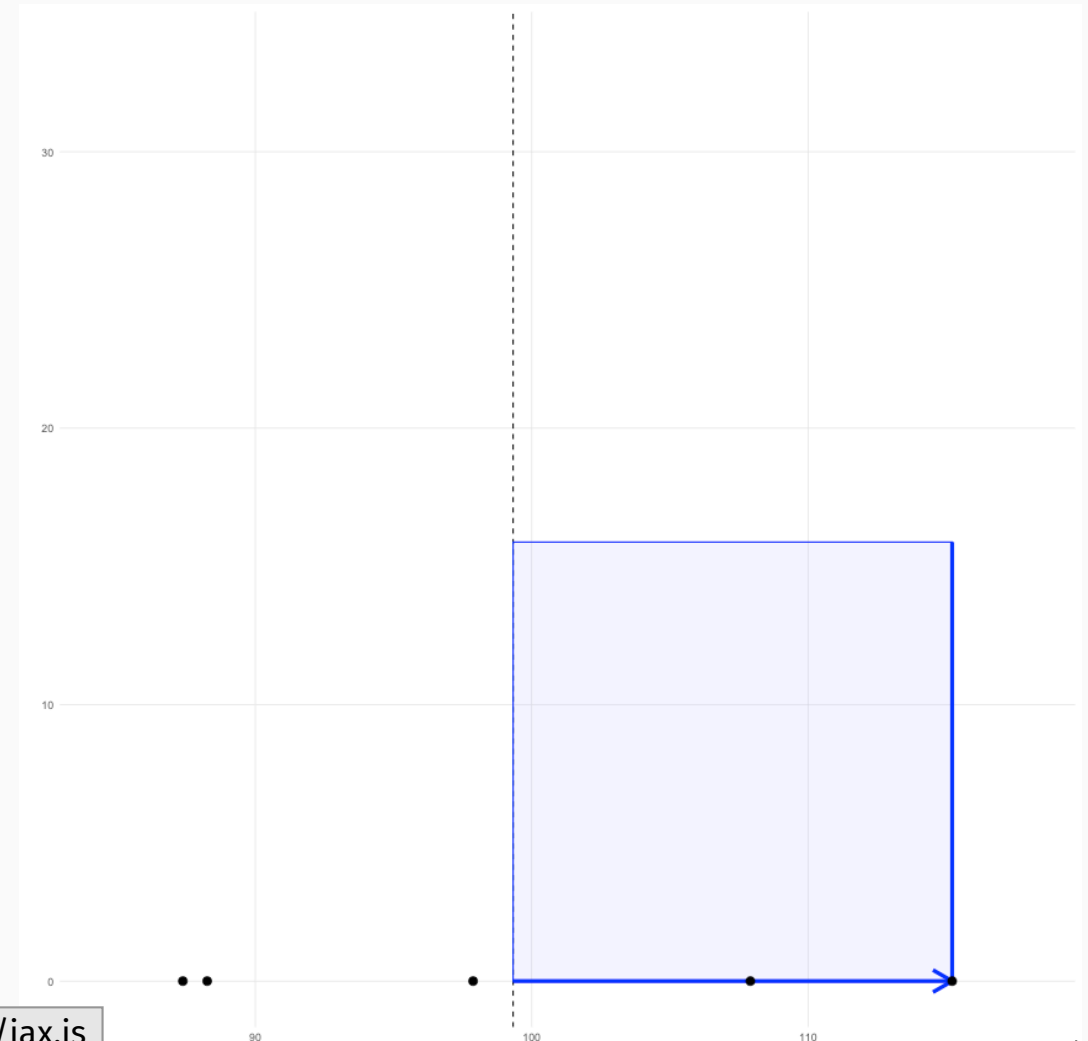
In the numerator, we square each of these deviances. We can conceptualize this as a square. Here, we add the deviance in the y direction.



# Variance (cont.)

Population Variance: *[Math Processing Error]*

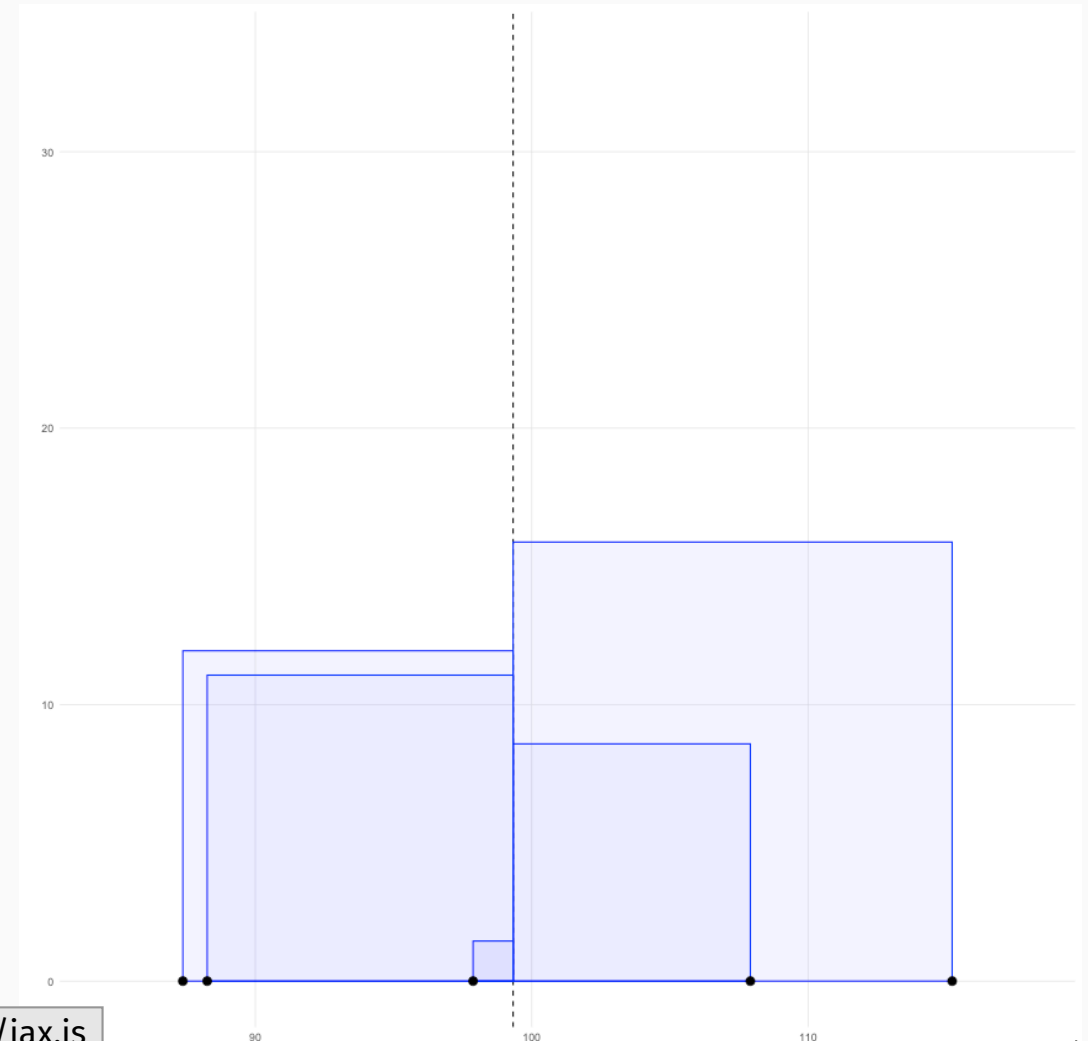
We end up with a square.



# Variance (cont.)

Population Variance: *[Math Processing Error]*

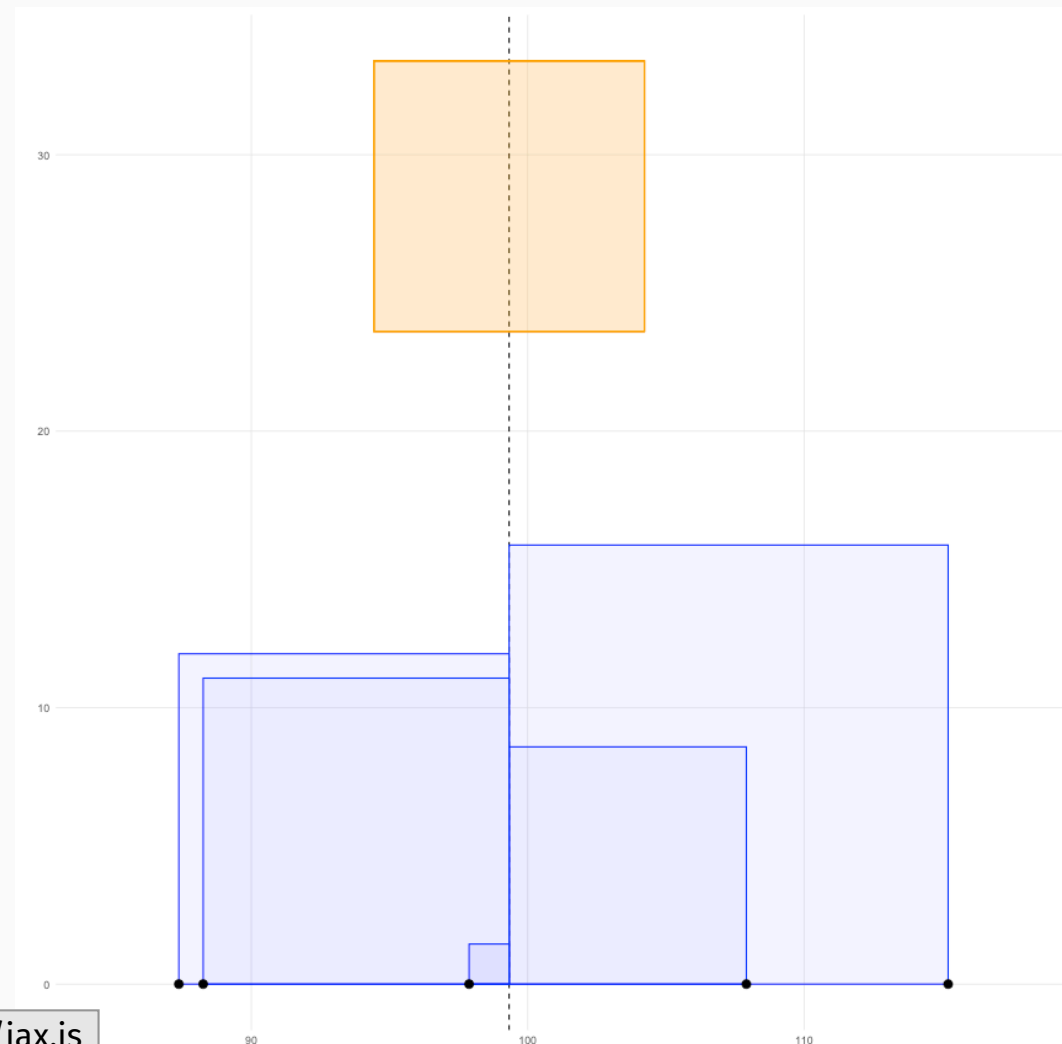
We can plot the squared deviance for all the data points. That is, each component in the numerator is the area of each of these squares.



# Variance (cont.)

Population Variance: *[Math Processing Error]*

The variance is therefore the average of the area of all these squares, here represented by the orange square.

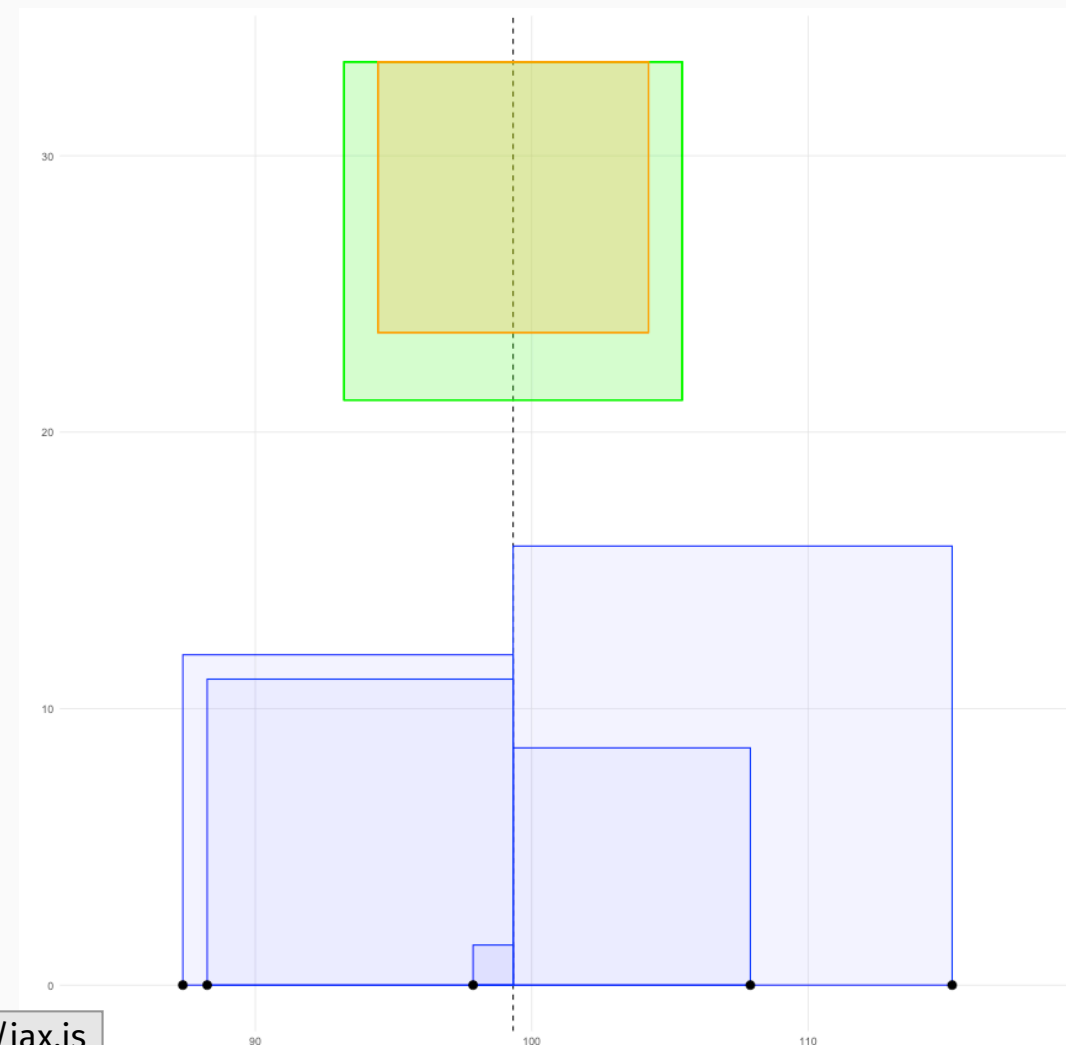


# Population versus Sample Variance

Typically we want the sample variance. The difference is we divide by *[Math Processing Error]* to calculate the sample variance. This results in a slightly larger area (variance) then if we divide by *[Math Processing Error]*.

Population Variance (yellow): *[Math Processing Error]*

Sample Variance (green): *[Math Processing Error]*



# Robust Statistics

Consider the following data randomly selected from the normal distribution:

```
set.seed(41)
x <- rnorm(30, mean = 100, sd = 15)
mean(x); sd(x)
```

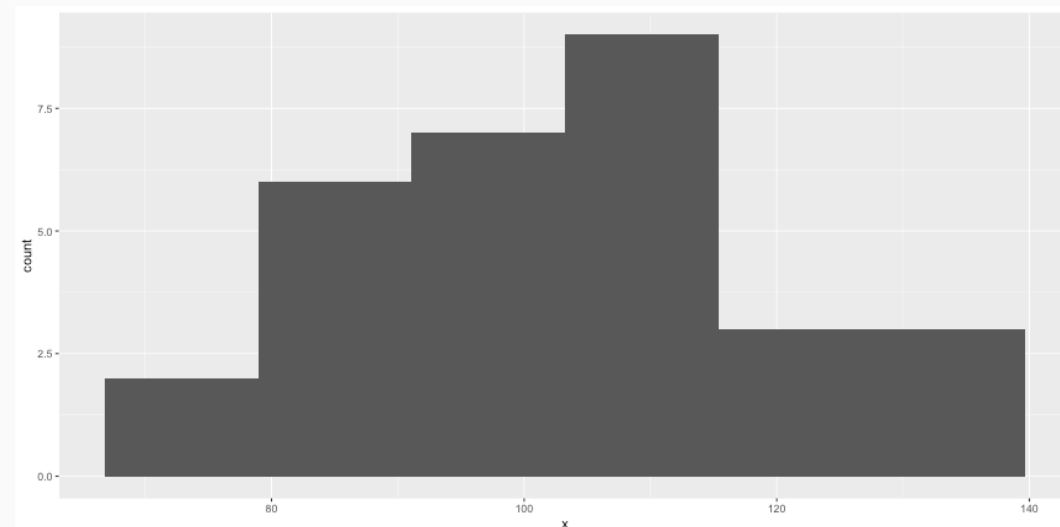
```
## [1] 103.1934
```

```
## [1] 16.8945
```

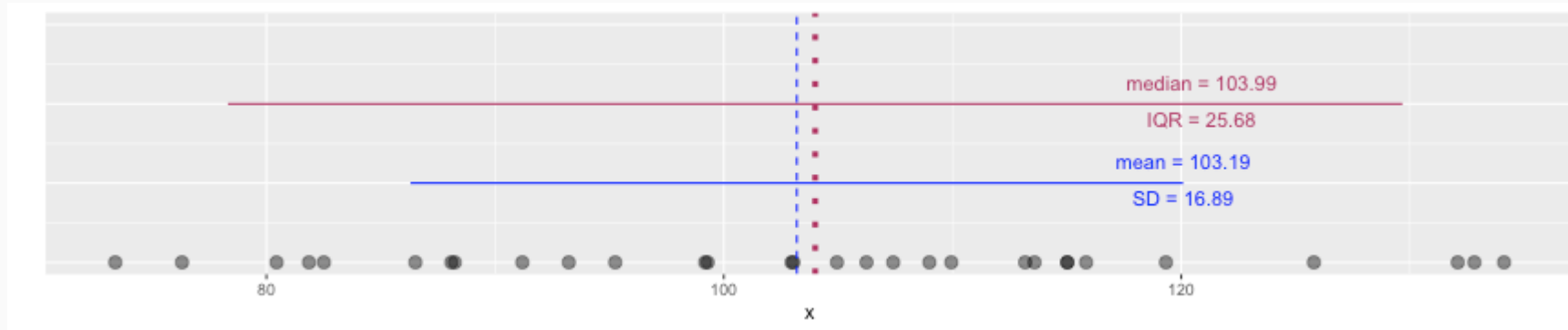
```
median(x); IQR(x)
```

```
## [1] 103.9947
```

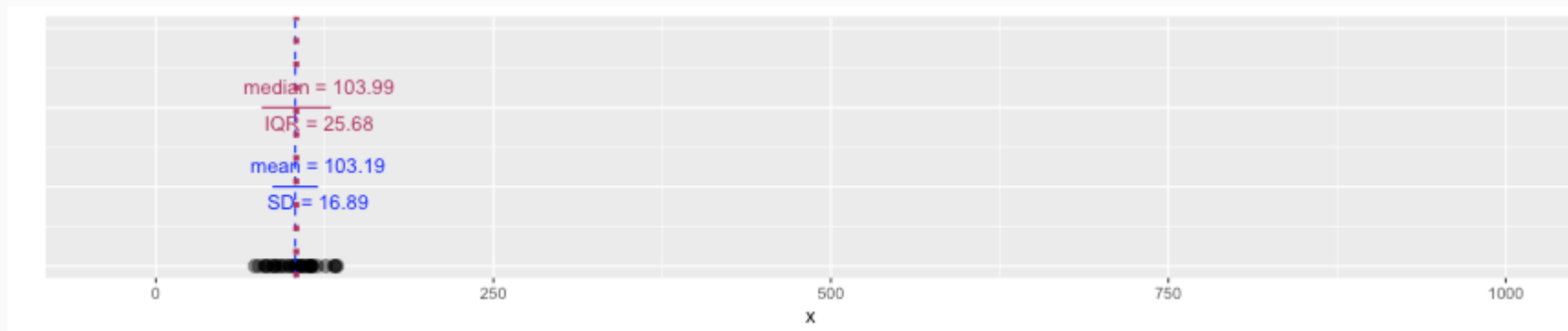
```
## [1] 25.68004
```



# Robust Statistics

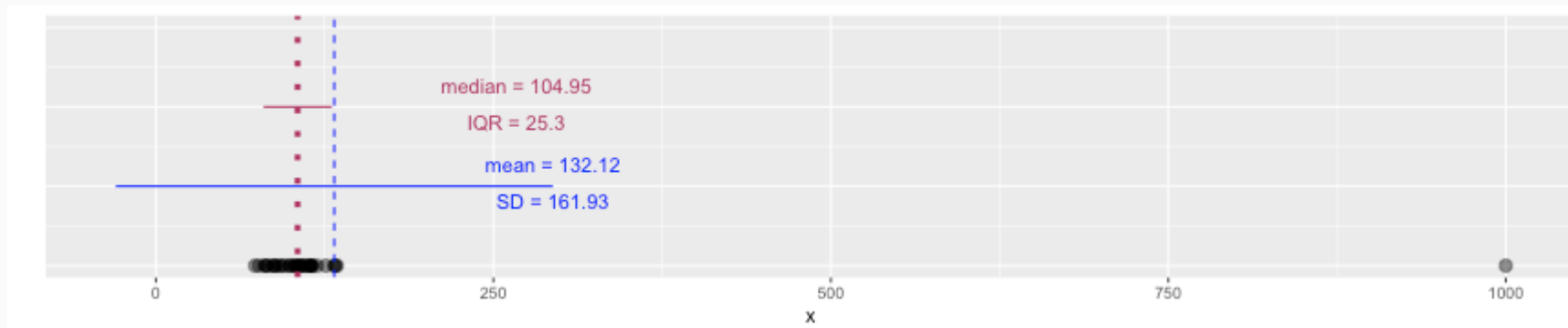


# Robust Statistics



Let's add an extreme value:

```
x <- c(x, 1000)
```



# Robust Statistics

Median and IQR are more robust to skewness and outliers than mean and SD. Therefore,

- for skewed distributions it is often more helpful to use median and IQR to describe the center and spread
- for symmetric distributions it is often more helpful to use the mean and SD to describe the center and spread

# Grammar of Graphics



# Data Visualizations with ggplot2



- `ggplot2` is an R package that provides an alternative framework based upon Wilkinson's (2005) Grammar of Graphics.
- `ggplot2` is, in general, more flexible for creating "prettier" and complex plots.
- Works by creating layers of different types of objects/geometries (i.e. bars, points, lines, polygons, etc.) `ggplot2` has at least three ways of creating plots:
  1. `qplot`
  2. `ggplot(...) + geom_XXX(...) + ...`
  3. `ggplot(...) + layer(...)`
- We will focus only on the second.

# Parts of a ggplot2 Statement



- Data

```
ggplot(myDataFrame, aes(x=x, y=y))
```

- Layers

```
geom_point(), geom_histogram()
```

- Facets

```
facet_wrap(~ cut), facet_grid(~ cut)
```

- Scales

```
scale_y_log10()
```

- Other options

```
ggtitle('my title'), ylim(c(0, 10000)), xlab('x-axis label')
```

# Lots of geoms



```
ls('package:ggplot2')[grep('^geom_', ls('package:ggplot2'))]
```

```
## [1] "geom_abline"      "geom_area"        "geom_bar"
## [4] "geom_bin_2d"      "geom_bin2d"       "geom_blank"
## [7] "geom_boxplot"     "geom_col"         "geom_contour"
## [10] "geom_contour_filled" "geom_count"      "geom_crossbar"
## [13] "geom_curve"       "geom_density"     "geom_density_2d"
## [16] "geom_density_2d_filled" "geom_density2d"  "geom_density2d_filled"
## [19] "geom_dotplot"     "geom_errorbar"    "geom_errorbarh"
## [22] "geom_freqpoly"    "geom_function"    "geom_hex"
## [25] "geom_histogram"   "geom_hline"       "geom_jitter"
## [28] "geom_label"       "geom_line"        "geom_linerange"
## [31] "geom_map"         "geom_path"        "geom_point"
## [34] "geom_pointrange"  "geom_polygon"     "geom_qq"
## [37] "geom_qq_line"     "geom_quantile"    "geom_raster"
## [40] "geom_rect"        "geom_ribbon"      "geom_rug"
## [43] "geom_segment"     "geom_sf"          "geom_sf_label"
## [46] "geom_sf_text"     "geom_smooth"      "geom_spoke"
## [49] "geom_step"        "geom_text"        "geom_tile"
## [52] "geom_violin"      "geom_vline"
```

# Data Visualization Cheat Sheet



## Data Visualization with ggplot2 : : CHEAT SHEET

### Basics

ggplot2 is based on the **grammar of graphics**, the idea that you can build every graph from the same components: a **data set**, a **coordinate system**, and **geoms**—visual marks that represent data points.



To display values, map variables in the data to visual properties of the geom (**aesthetics**) like **size**, **color**, and **x** and **y** locations.



Complete the template below to build a graph.

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>)) +  
  stat = <STAT>, position = <POSITION> +  
  <COORDINATE_FUNCTION> +  
  <FACET_FUNCTION> +  
  <SCALE_FUNCTION> +  
  <THEME_FUNCTION>
```

ggplot(data = mpg, aes(x = cty, y = hwy)) Begins a plot that you finish by adding layers to. Add one geom function per layer.

ggplot(x = cty, y = hwy, data = mpg, geom = "point") Creates a complete plot with given data, geom, and mappings. Supplies many useful defaults.

last\_plot() Returns the last plot

ggsave("plot.png", width = 5, height = 5) Saves last plot as 5' x 5' file named "plot.png" in working directory. Matches file type to file extension.

### Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

#### GRAPHICAL PRIMITIVES

```
a <- ggplot(economics, aes(date, unemployment))  
b <- ggplot(seals, aes(x = long, y = lat))  
  
a + geom_blank()  
(Useful for expanding limits)  
  
b + geom_curve(aes(yend = lat + 1,  
  xend = long + 1, curvature = z)) - x, yend, y, yend,  
  alpha, angle, color, curvature, linetype, size  
  
a + geom_path(linetype = "butt", linejoin = "round",  
  linemitre = 1)  
x, y, alpha, color, group, linetype, size  
  
a + geom_polygon(aes(group = group))  
x, y, alpha, color, fill, group, linetype, size  
  
b + geom_rect(aes(xmin = long, ymin = lat, xmax =  
  long + 1, ymax = lat + 1)) - xmax, xmin, ymax,  
  ymin, alpha, color, fill, linetype, size  
  
a + geom_ribbon(aes(ymin = unemployment - 900,  
  ymax = unemployment + 900)) - x, ymax, ymin,  
  alpha, color, fill, group, linetype, size
```

#### LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

```
b + geom_abline(aes(intercept = 0, slope = 1))  
b + geom_hline(aes(yintercept = lat))  
b + geom_vline(aes(xintercept = long))  
  
b + geom_segment(aes(yend = lat + 1, xend = long + 1))  
b + geom_spoke(aes(angle = 1:1155, radius = 1))
```

#### ONE VARIABLE continuous

```
c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)  
  
c + geom_area(stat = "bin")  
x, y, alpha, color, fill, linetype, size  
  
c + geom_density(kernel = "gaussian")  
x, y, alpha, color, fill, group, linetype, size, weight  
  
c + geom_dotplot()  
x, y, alpha, color, fill  
  
c + geom_freqpoly() x, y, alpha, color, group,  
  linetype, size  
  
c + geom_histogram(binwidth = 5) x, y, alpha,  
  color, fill, linetype, size, weight  
  
c2 + geom_qq(aes(sample = hwy)) x, y, alpha,  
  color, fill, linetype, size, weight
```

#### discrete

```
d <- ggplot(mpg, aes(fill))  
  
d + geom_bar()  
x, alpha, color, fill, linetype, size, weight
```

#### TWO VARIABLES

continuous x, continuous y

```
e <- ggplot(mpg, aes(cty, hwy))  
  
e + geom_label(aes(label = cty), nudge_x = 1,  
  nudge_y = 1, check_overlap = TRUE) x, y, label,  
  alpha, angle, color, family, fontface, hjust,  
  lineheight, size, vjust  
  
e + geom_jitter(height = 2, width = 2)  
x, y, alpha, color, fill, shape, size  
  
e + geom_point(), x, y, alpha, color, fill, shape,  
  size, stroke  
  
e + geom_quantile(), x, y, alpha, color, group,  
  linetype, size, weight  
  
e + geom_rug(sides = "bl"), x, y, alpha, color,  
  linetype, size  
  
e + geom_smooth(method = lm), x, y, alpha, color,  
  fill, group, linetype, size, weight  
  
e + geom_text(aes(label = cty), nudge_x = 1,  
  nudge_y = 1, check_overlap = TRUE) x, y, label,  
  alpha, angle, color, family, fontface, hjust,  
  lineheight, size, vjust
```

discrete x, continuous y

```
f <- ggplot(mpg, aes(class, hwy))  
  
f + geom_col(), x, y, alpha, color, fill, group,  
  linetype, size  
  
f + geom_boxplot(), x, y, lower, middle, upper,  
  ymax, ymin, alpha, color, fill, group, linetype,  
  shape, size, weight  
  
f + geom_dotplot(binaxis = "y", stackdir =  
  "center"), x, y, alpha, color, fill, group  
  
f + geom_violin(scale = "area"), x, y, alpha, color,  
  fill, group, linetype, size, weight
```

discrete x, discrete y

```
g <- ggplot(diamonds, aes(cut, color))  
  
g + geom_count(), x, y, alpha, color, fill, shape,  
  size, stroke
```

#### THREE VARIABLES

```
seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2)) l <- ggplot(seals, aes(long, lat))  
  
l + geom_contour(aes(z = z))  
x, y, z, alpha, colour, group, linetype,  
  size, weight  
  
l + geom_raster(aes(fill = z), hjust = 0.5, vjust = 0.5,  
  interpolate = FALSE)  
x, y, alpha, fill  
  
l + geom_tile(aes(fill = z)), x, y, alpha, color, fill,  
  linetype, size, width
```

continuous bivariate distribution

```
h <- ggplot(diamonds, aes(carat, price))  
  
h + geom_bin2d(binwidth = c(0.25, 500))  
x, y, alpha, color, fill, linetype, size, weight  
  
h + geom_density2d()  
x, y, alpha, colour, group, linetype, size  
  
h + geom_hex()  
x, y, alpha, colour, fill, size
```

continuous function

```
i <- ggplot(economics, aes(date, unemployment))  
  
i + geom_area()  
x, y, alpha, color, fill, linetype, size  
  
i + geom_line()  
x, y, alpha, color, group, linetype, size  
  
i + geom_step(direction = "hv")  
x, y, alpha, color, group, linetype, size
```

visualizing error

```
df <- data.frame(grp = c("A", "B"), fit = 4.5, se = 1.2)  
j <- ggplot(df, aes(grp, fit, ymin = fit - se, ymax = fit + se))  
  
j + geom_crossbar(fatten = 2)  
x, y, ymax, ymin, alpha, color, fill, group, linetype,  
  size  
  
j + geom_errorbar(), x, ymax, ymin, alpha, color,  
  group, linetype, size, width (also  
  geom_errorbarh())  
  
j + geom_linerange()  
x, ymin, ymax, alpha, color, group, linetype, size  
  
j + geom_pointrange()  
x, y, ymin, ymax, alpha, color, fill, group, linetype,  
  shape, size
```

maps

```
data <- data.frame(murder = USArrests$Murder,  
  state = tolower(rownames(USArrests)))  
map <- map_data("state")  
k <- ggplot(data, aes(fill = murder))  
  
k + geom_map(aes(map_id = state), map = map)  
+ expand_limits(x = map$long, y = map$lat),  
  map_id, alpha, color, fill, linetype, size
```

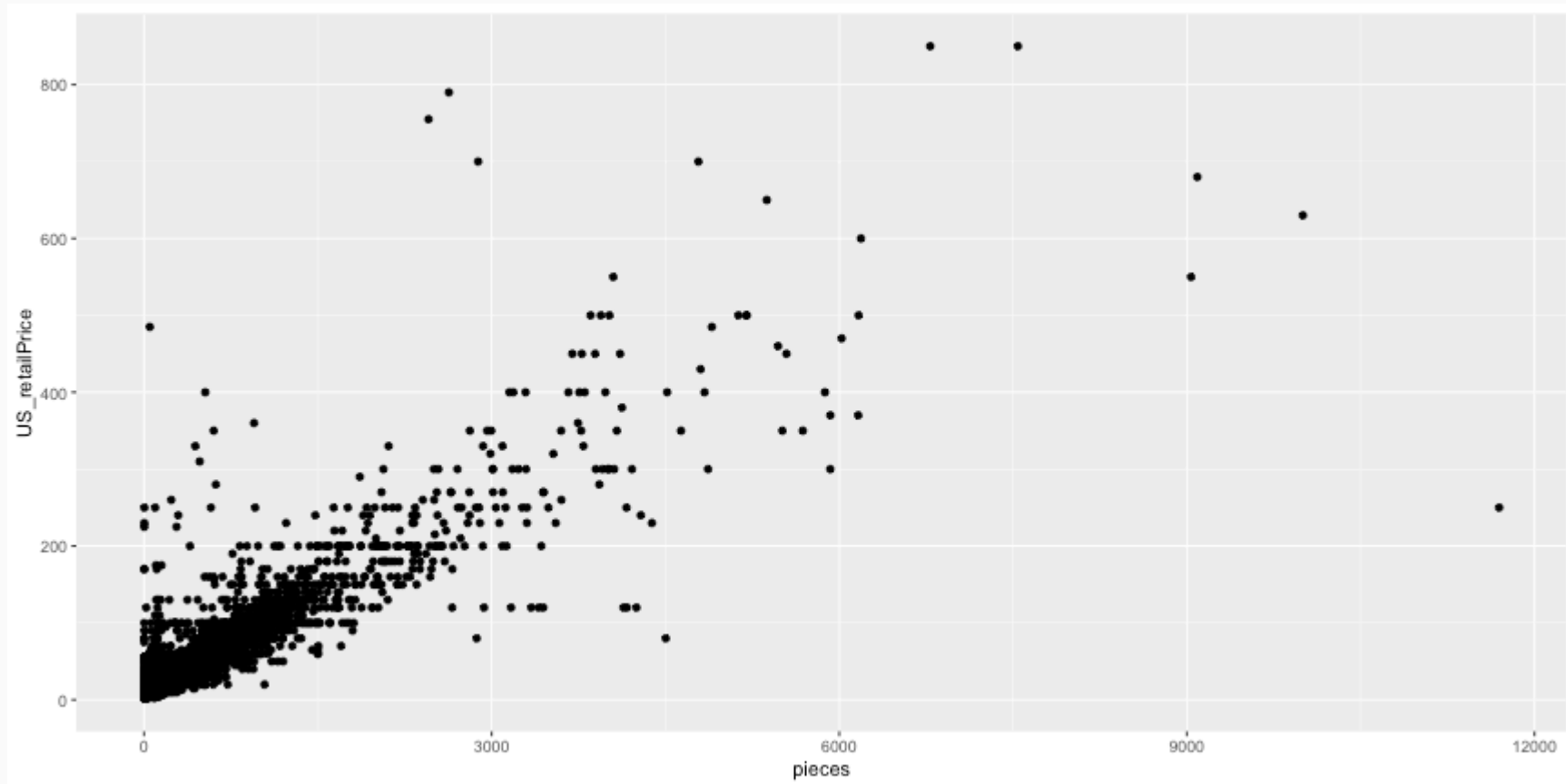


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# Scatterplot



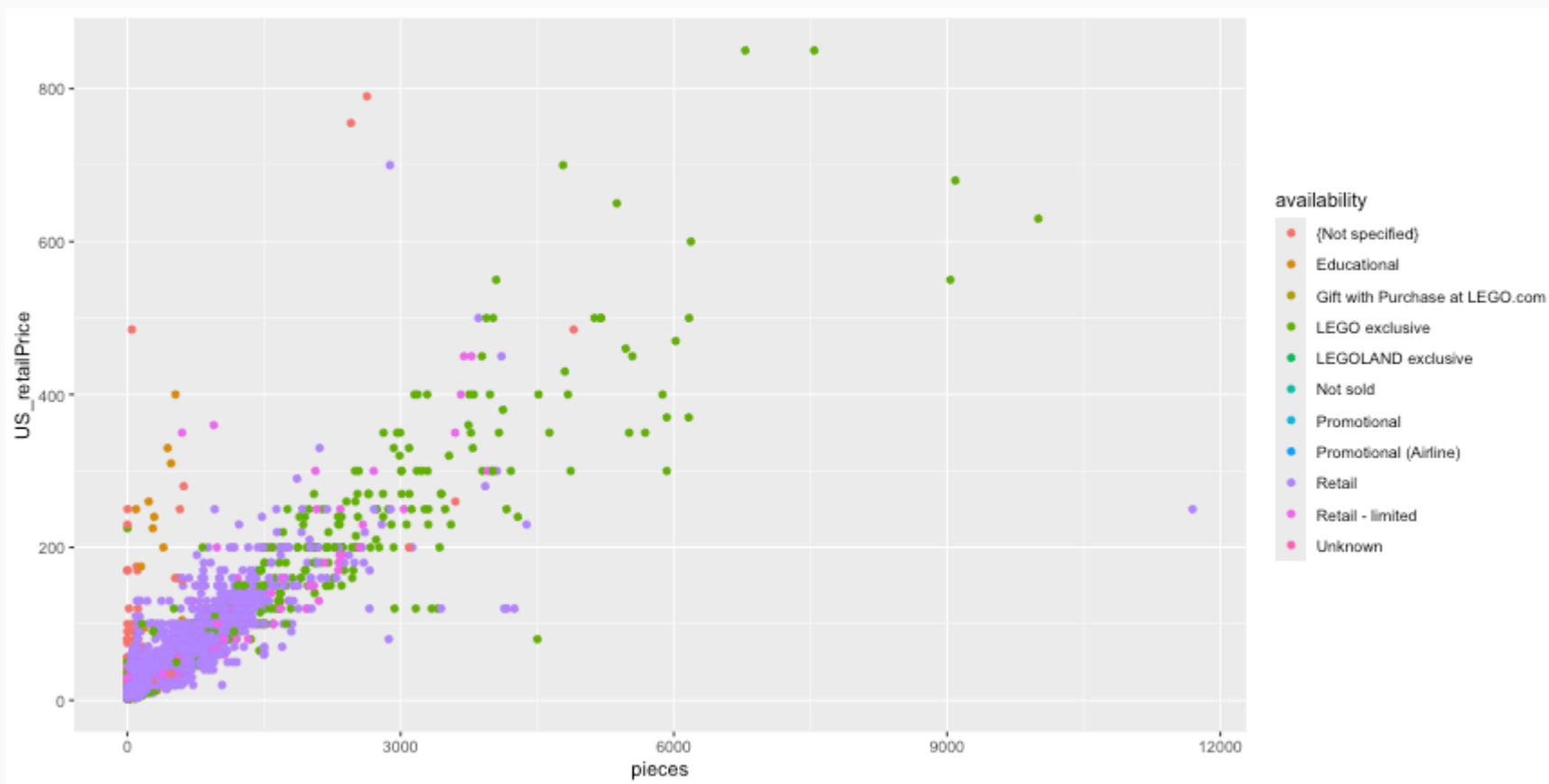
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice)) + geom_point()
```



# Scatterplot (cont.)



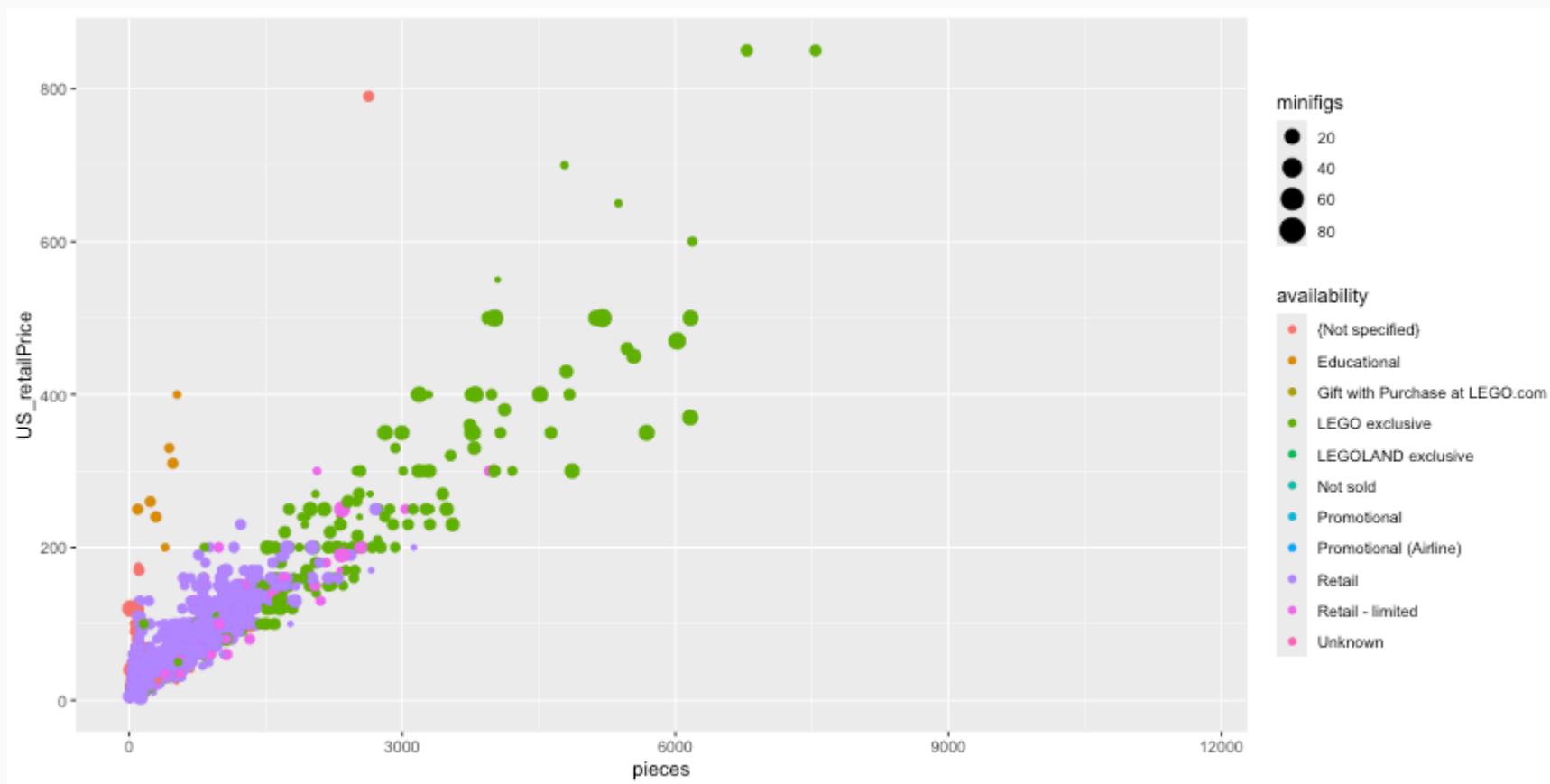
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice, color=availability)) + geom_point()
```



# Scatterplot (cont.)



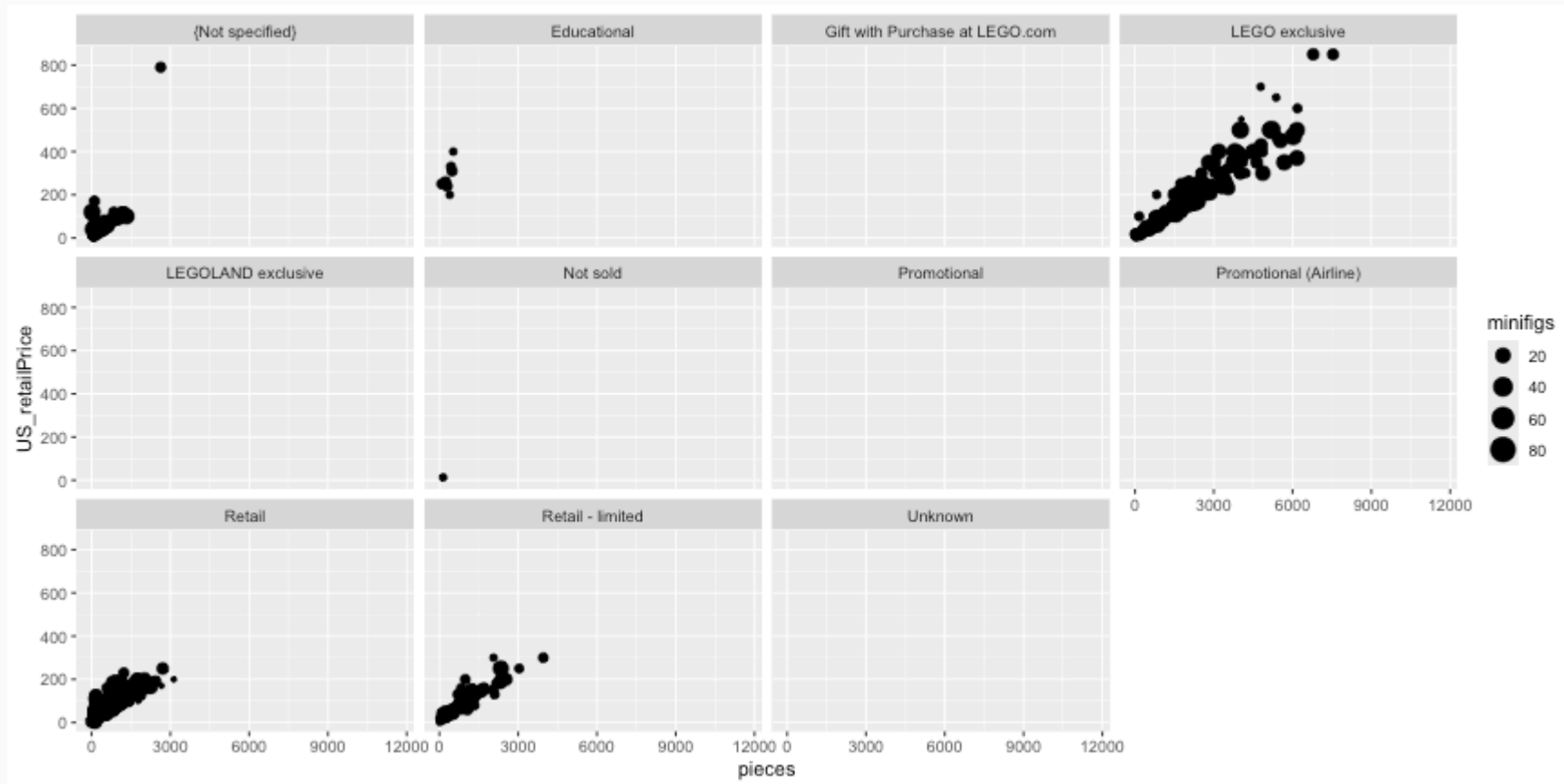
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice, size=minifigs, color=availability)) + geom_point()
```



# Scatterplot (cont.)



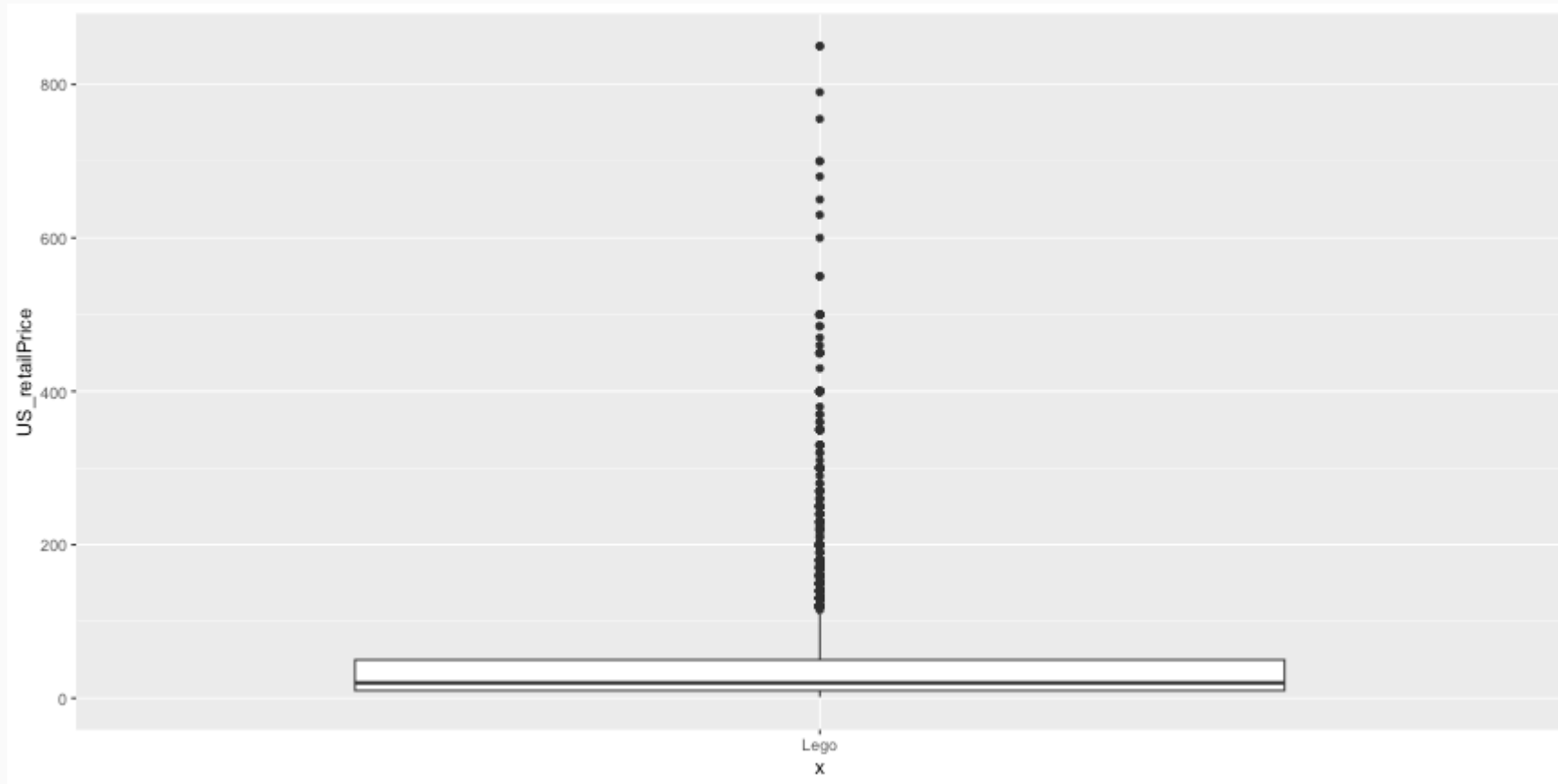
```
ggplot(legosets, aes(x=pieces, y=US_retailPrice, size=minifigs)) + geom_point() + facet_wrap(~ availability)
```



# Boxplots



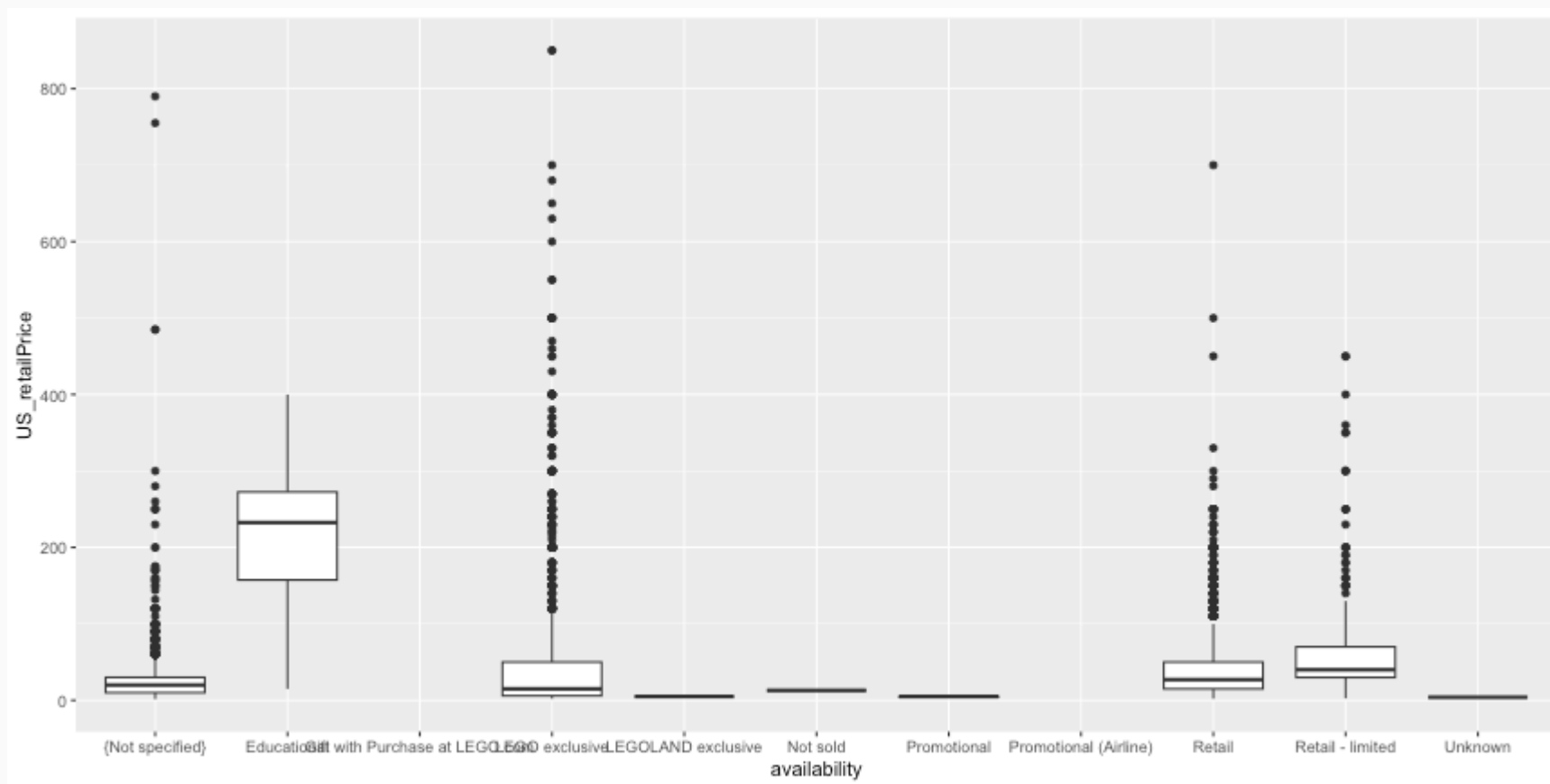
```
ggplot(legosets, aes(x='Lego', y=US_retailPrice)) + geom_boxplot()
```



# Boxplots (cont.)

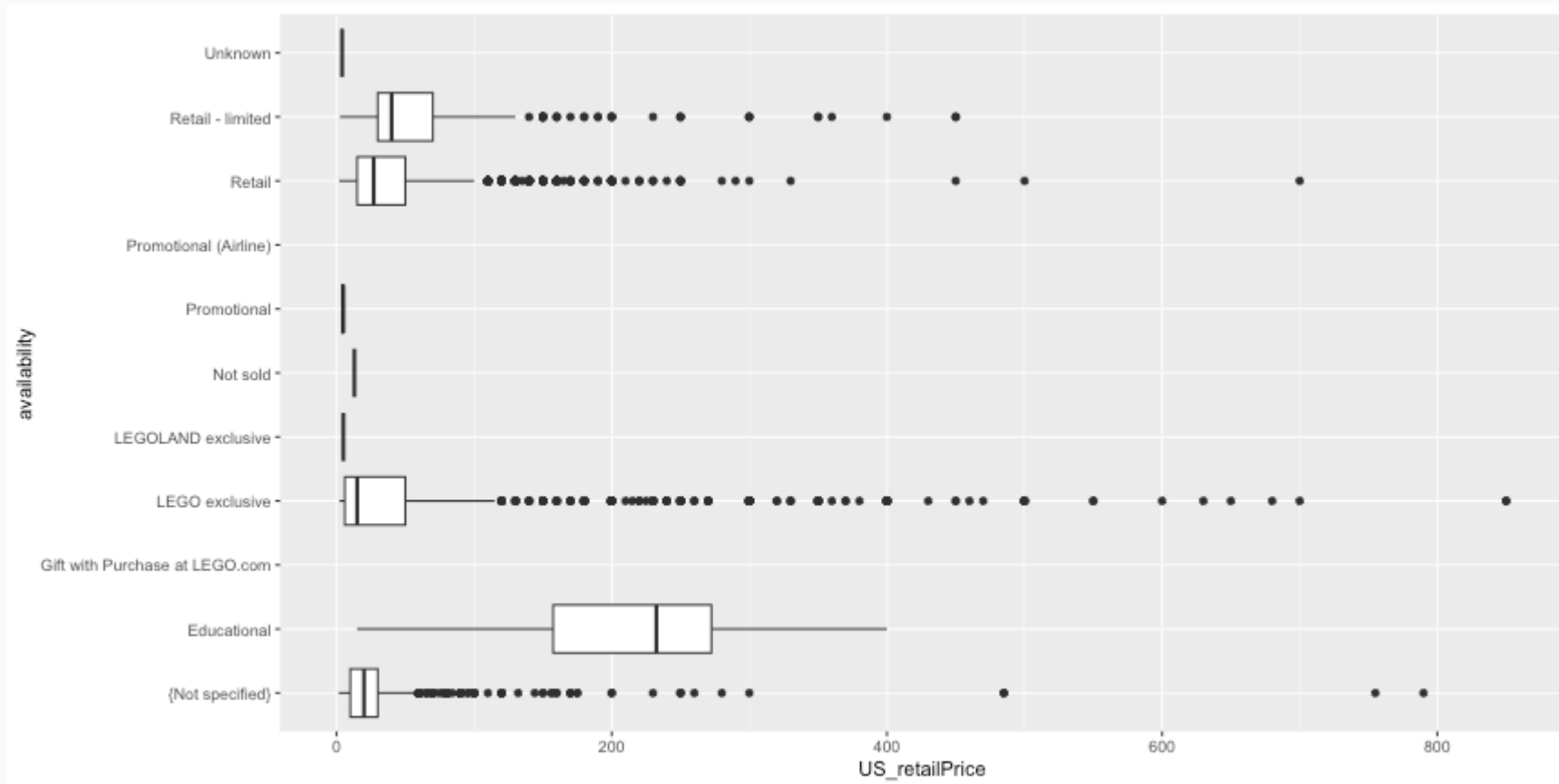


```
ggplot(legosets, aes(x=availability, y=US_retailPrice)) + geom_boxplot()
```





```
ggplot(legosets, aes(x=availability, y=US_retailPrice)) + geom_boxplot() + coord_flip()
```

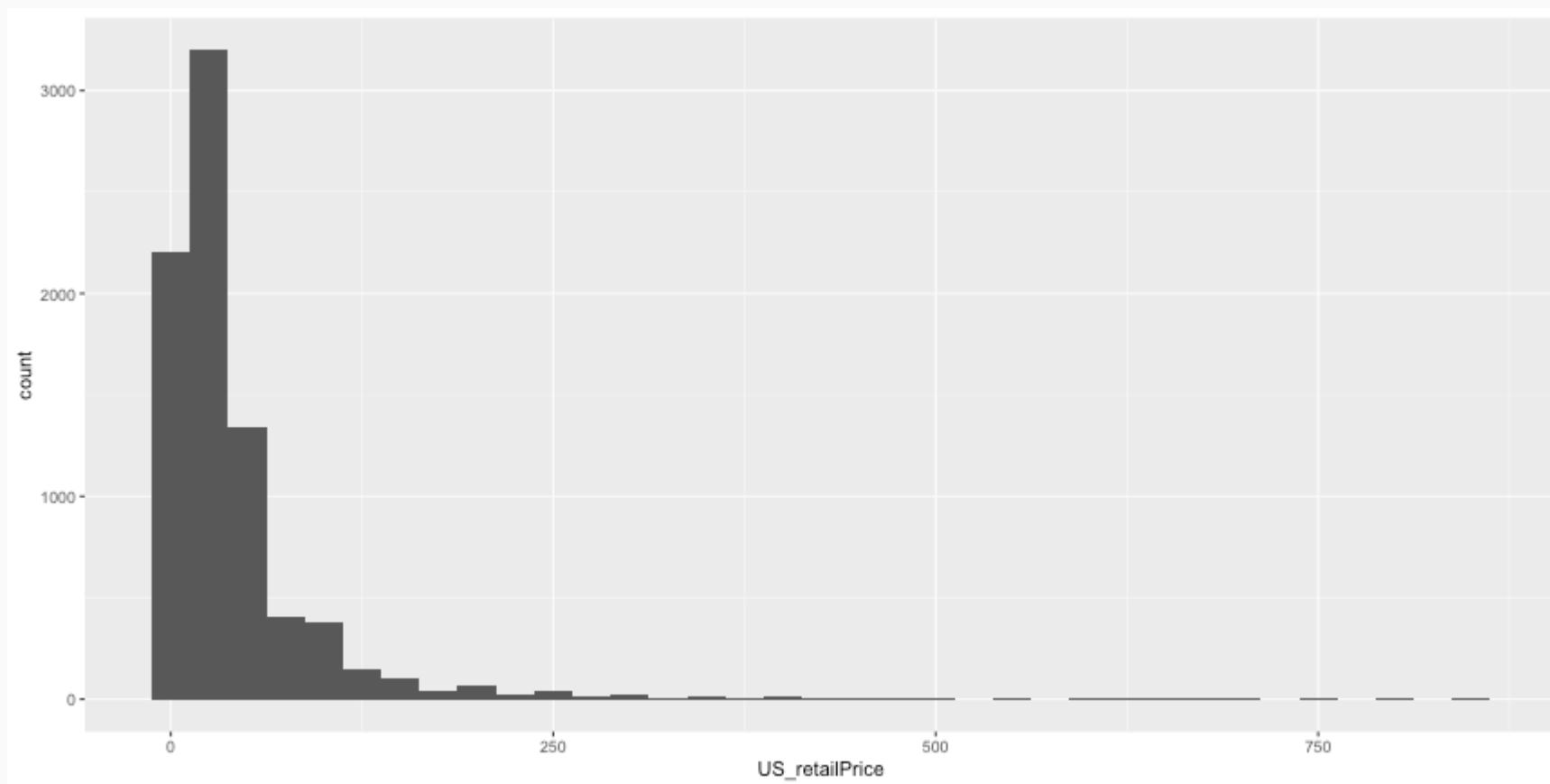


File failed to load: <http://mathjax.rstudio.com/latest/jax/output/CommonHTML/jax.js>

# Histograms



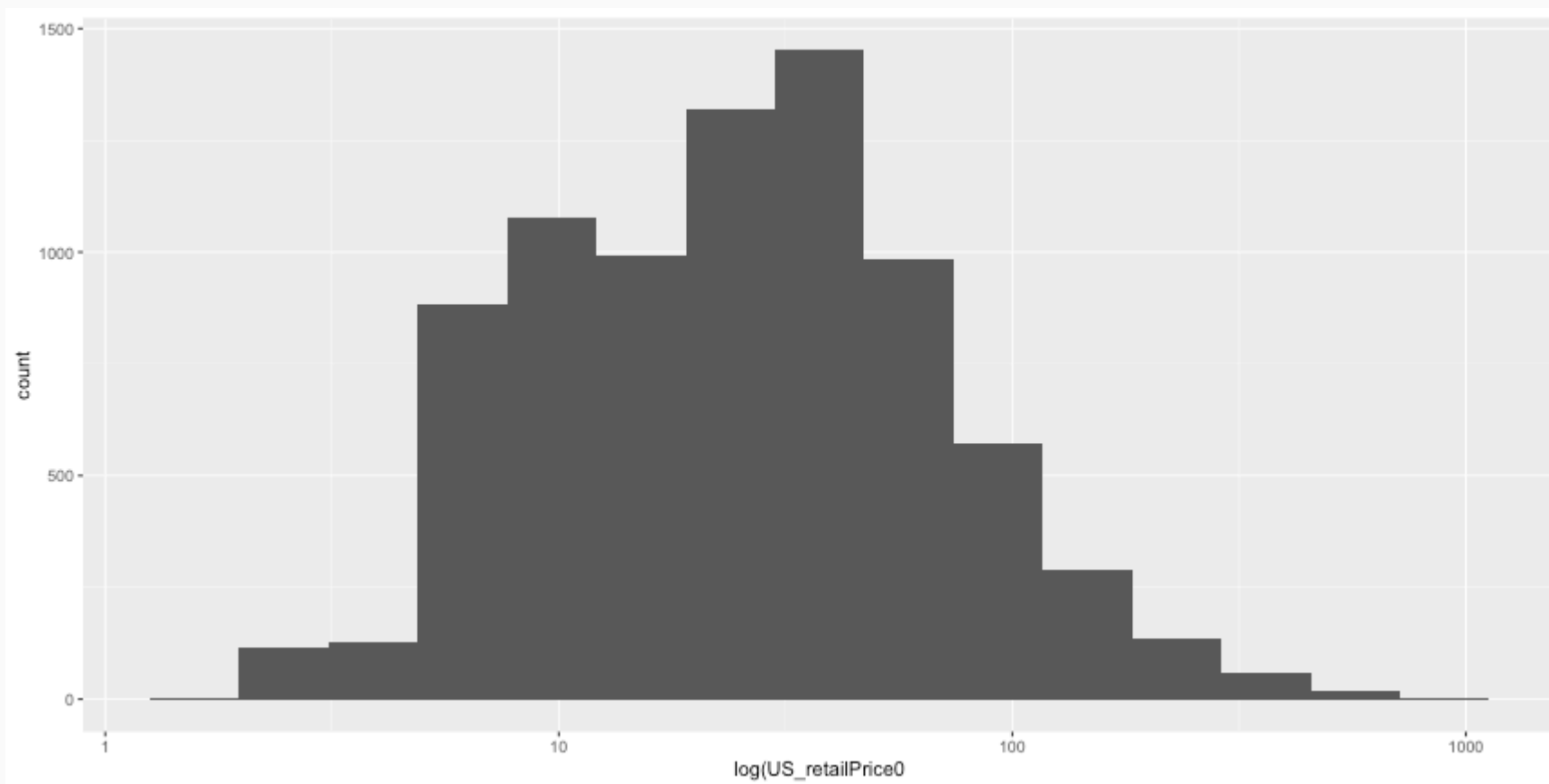
```
ggplot(legosets, aes(x = US_retailPrice)) + geom_histogram(binwidth = 25)
```



# Histograms (cont.)



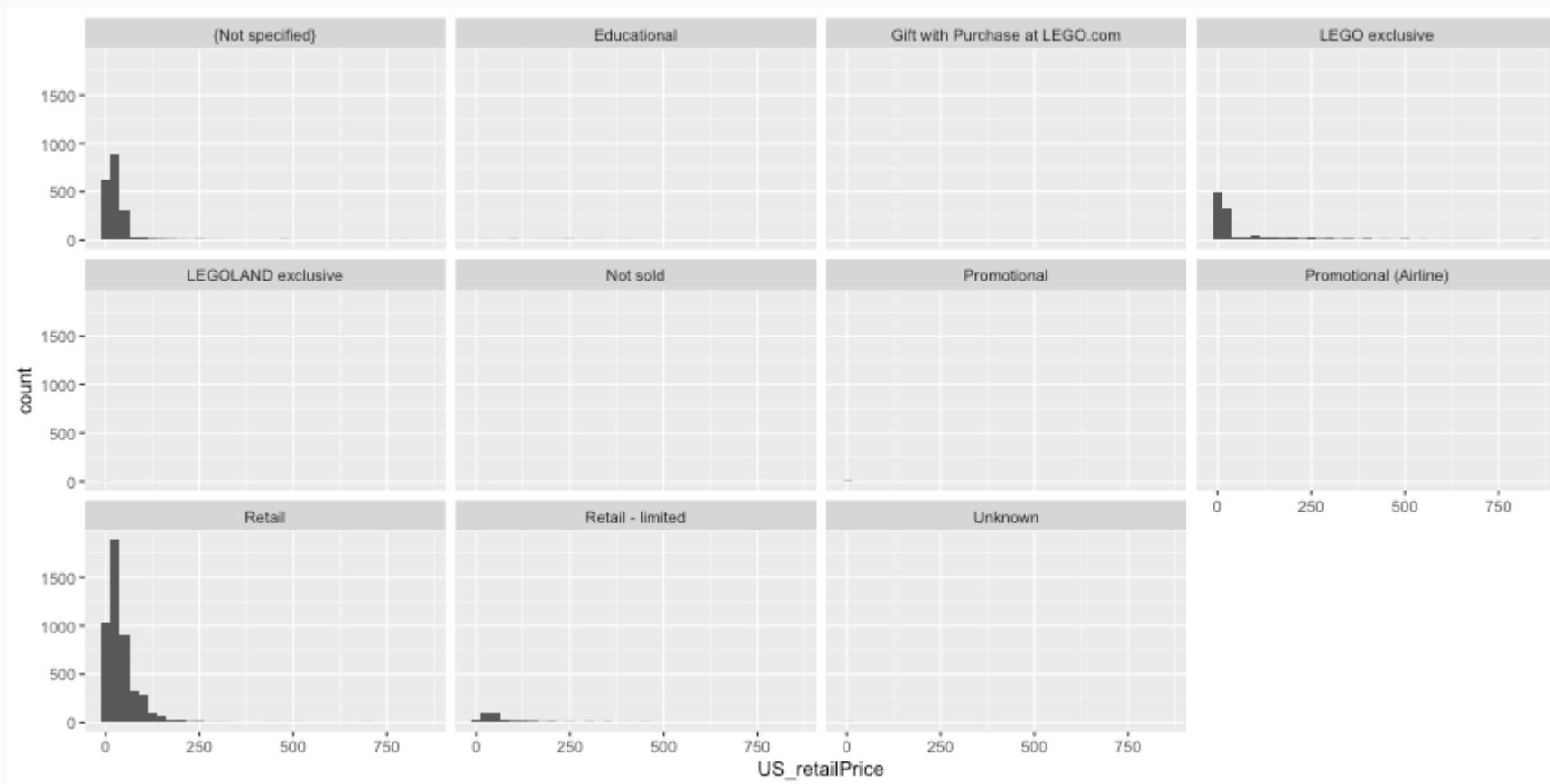
```
ggplot(legosets, aes(x = US_retailPrice)) + geom_histogram(bins = 15) + scale_x_log10() + xlab('log(US_retailPrice0)')
```



# Histograms (cont.)



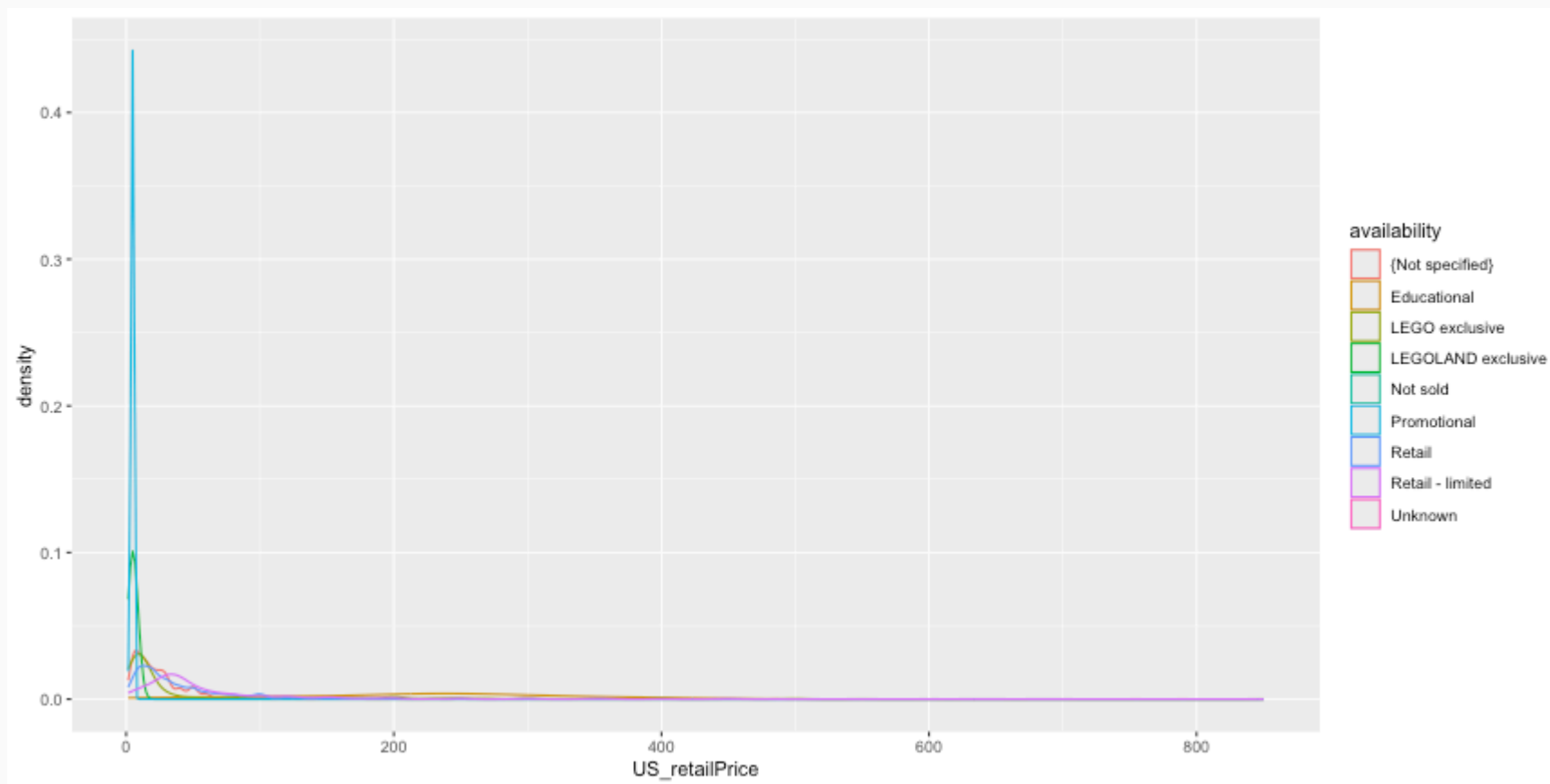
```
ggplot(legosets, aes(x = US_retailPrice)) + geom_histogram(binwidth = 25) + facet_wrap(~ availability)
```



# Density Plots



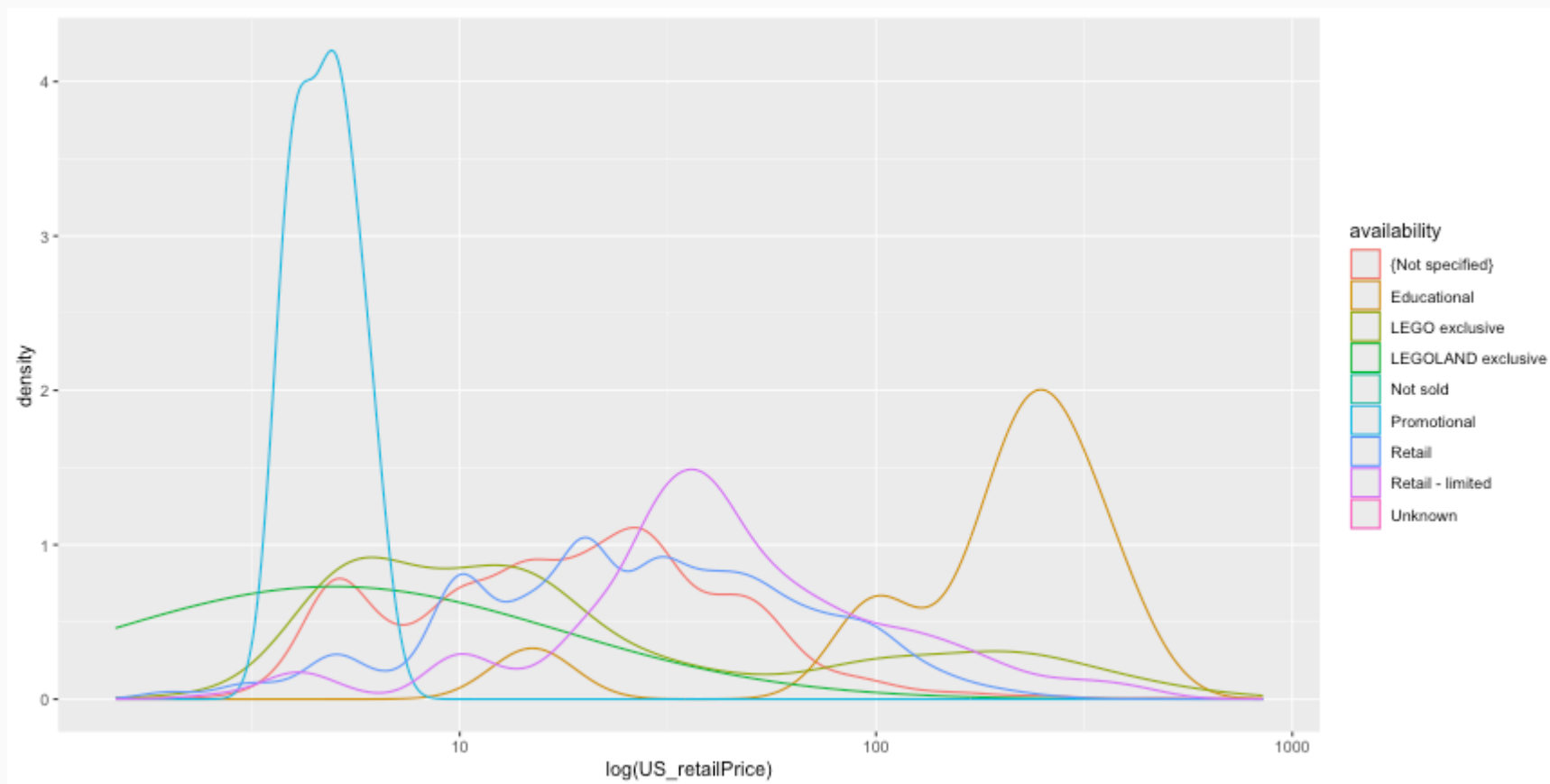
```
ggplot(legosets, aes(x = US_retailPrice, color = availability)) + geom_density()
```



# Density Plots (cont.)



```
ggplot(legosets, aes(x = US_retailPrice, color = availability)) + geom_density() + scale_x_log10() + xlab('log(US_ret
```



## ggplot2 aesthetics cheat sheet

Use this table to find the right aesthetics for your geoms:

Aesthetics that usually must be mapped to the data: use inside `aes()`

Aesthetics that can be mapped to the data: use in or outside `aes()`

Aesthetics that cannot be mapped to the data: use outside `aes()`

e.g., `ggplot(mpg, aes(x = class, y = displ)) + geom_col(aes(fill = class), width = .9)`

|                           | color | linetype | fill | y    | xmax | ymax | yend   | shape  | width  | angle  | hjust | label  | fontface   |                           |
|---------------------------|-------|----------|------|------|------|------|--------|--------|--------|--------|-------|--------|------------|---------------------------|
| group                     | size  | alpha    | x    | xmin | ymin | xend | weight | stroke | height | radius | vjust | family | lineheight |                           |
| area                      |       |          |      |      |      |      |        |        |        |        |       |        |            | area                      |
| bar (vertical)            |       |          |      |      |      |      |        |        |        |        |       |        |            | bar (vertical)            |
| bar (horizontal)          |       |          |      |      |      |      |        |        |        |        |       |        |            | bar (horizontal)          |
| bin2d                     |       |          |      |      |      |      |        |        |        |        |       |        |            | bin2d                     |
| boxplot                   |       |          |      |      |      |      |        |        |        |        |       |        |            | boxplot                   |
| col                       |       |          |      |      |      |      |        |        |        |        |       |        |            | col                       |
| contour                   |       |          |      |      |      |      |        |        |        |        |       |        |            | contour                   |
| contour_filled            |       |          |      |      |      |      |        |        |        |        |       |        |            | contour_filled            |
| count                     |       |          |      |      |      |      |        |        |        |        |       |        |            | count                     |
| crossbar (vertical)       |       |          |      |      |      |      |        |        |        |        |       |        |            | crossbar (vertical)       |
| crossbar (horizontal)     |       |          |      |      |      |      |        |        |        |        |       |        |            | crossbar (horizontal)     |
| curve                     |       |          |      |      |      |      |        |        |        |        |       |        |            | curve                     |
| density                   |       |          |      |      |      |      |        |        |        |        |       |        |            | density                   |
| density_2d                |       |          |      |      |      |      |        |        |        |        |       |        |            | density_2d                |
| dotplot                   |       |          |      |      |      |      |        |        |        |        |       |        |            | dotplot                   |
| errorbar                  |       |          |      |      |      |      |        |        |        |        |       |        |            | errorbar                  |
| errorbarh                 |       |          |      |      |      |      |        |        |        |        |       |        |            | errorbarh                 |
| freqpoly                  |       |          |      |      |      |      |        |        |        |        |       |        |            | freqpoly                  |
| hex                       |       |          |      |      |      |      |        |        |        |        |       |        |            | hex                       |
| histogram (on x-axis)     |       |          |      |      |      |      |        |        |        |        |       |        |            | histogram (on x-axis)     |
| histogram (on y-axis)     |       |          |      |      |      |      |        |        |        |        |       |        |            | histogram (on y-axis)     |
| jitter                    |       |          |      |      |      |      |        |        |        |        |       |        |            | jitter                    |
| label                     |       |          |      |      |      |      |        |        |        |        |       |        |            | label                     |
| line                      |       |          |      |      |      |      |        |        |        |        |       |        |            | line                      |
| linrange (vertical)       |       |          |      |      |      |      |        |        |        |        |       |        |            | linrange (vertical)       |
| linrange (horizontal)     |       |          |      |      |      |      |        |        |        |        |       |        |            | linrange (horizontal)     |
| map                       |       |          |      |      |      |      |        |        |        |        |       |        |            | map                       |
| path                      |       |          |      |      |      |      |        |        |        |        |       |        |            | path                      |
| point                     |       |          |      |      |      |      |        |        |        |        |       |        |            | point                     |
| pointrange (vertical)     |       |          |      |      |      |      |        |        |        |        |       |        |            | pointrange (vertical)     |
| pointrange (horizontal)   |       |          |      |      |      |      |        |        |        |        |       |        |            | pointrange (horizontal)   |
| polygon                   |       |          |      |      |      |      |        |        |        |        |       |        |            | polygon                   |
| quantile                  |       |          |      |      |      |      |        |        |        |        |       |        |            | quantile                  |
| raster                    |       |          |      |      |      |      |        |        |        |        |       |        |            | raster                    |
| rect                      |       |          |      |      |      |      |        |        |        |        |       |        |            | rect                      |
| ribbon (variation y-axis) |       |          |      |      |      |      |        |        |        |        |       |        |            | ribbon (variation y-axis) |
| ribbon (variation x-axis) |       |          |      |      |      |      |        |        |        |        |       |        |            | ribbon (variation x-axis) |
| rug                       |       |          |      |      |      |      |        |        |        |        |       |        |            | rug                       |
| segment                   |       |          |      |      |      |      |        |        |        |        |       |        |            | segment                   |
| smooth                    |       |          |      |      |      |      |        |        |        |        |       |        |            | smooth                    |
| spoke                     |       |          |      |      |      |      |        |        |        |        |       |        |            | spoke                     |
| step                      |       |          |      |      |      |      |        |        |        |        |       |        |            | step                      |
| text                      |       |          |      |      |      |      |        |        |        |        |       |        |            | text                      |
| tile                      |       |          |      |      |      |      |        |        |        |        |       |        |            | tile                      |
| violin                    |       |          |      |      |      |      |        |        |        |        |       |        |            | violin                    |

● usually must be inside `aes()` ■ can be inside `aes()` ◆ must be outside `aes()`



Likert scales are a type of questionnaire where respondents are asked to rate items on scales usually ranging from four to seven levels (e.g. strongly disagree to strongly agree).

```
library(likert)
library(reshape)
data(pisaitems)
items24 <- pisaitems[,substr(names(pisaitems), 1,5) == 'ST24Q']
items24 <- rename(items24, c(
  ST24Q01="I read only if I have to.",
  ST24Q02="Reading is one of my favorite hobbies.",
  ST24Q03="I like talking about books with other people.",
  ST24Q04="I find it hard to finish books.",
  ST24Q05="I feel happy if I receive a book as a present.",
  ST24Q06="For me, reading is a waste of time.",
  ST24Q07="I enjoy going to a bookstore or a library.",
  ST24Q08="I read only to get information that I need.",
  ST24Q09="I cannot sit still and read for more than a few minutes.",
  ST24Q10="I like to express my opinions about books I have read.",
  ST24Q11="I like to exchange books with my friends."))
```



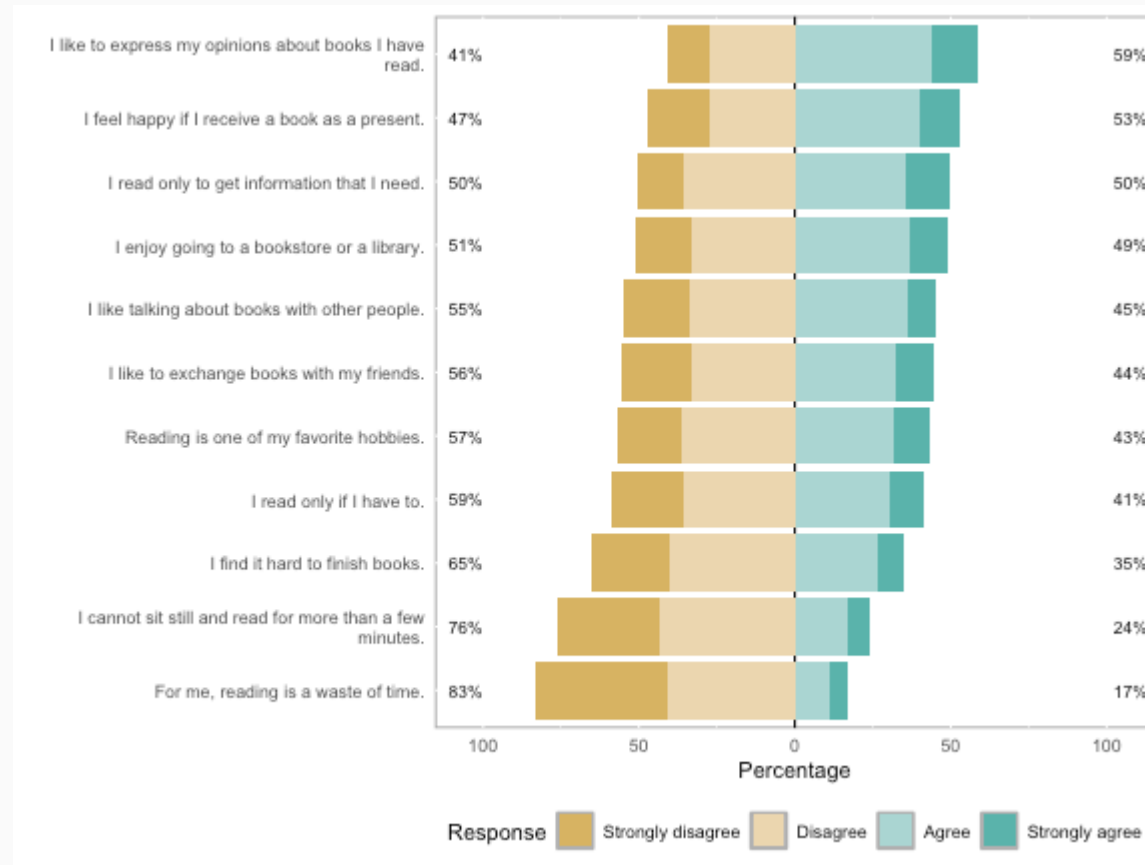
```
l24 <- likert(items24)
summary(l24)
```

```
##                               Item      low neutral
## 10  I like to express my opinions about books I have read. 41.07516      0
## 5   I feel happy if I receive a book as a present. 46.93475      0
## 8   I read only to get information that I need. 50.39874      0
## 7   I enjoy going to a bookstore or a library. 51.21231      0
## 3   I like talking about books with other people. 54.99129      0
## 11  I like to exchange books with my friends. 55.54115      0
## 2   Reading is one of my favorite hobbies. 56.64470      0
## 1   I read only if I have to. 58.72868      0
## 4   I find it hard to finish books. 65.35125      0
## 9  I cannot sit still and read for more than a few minutes. 76.24524      0
## 6   For me, reading is a waste of time. 82.88729      0
##      high      mean      sd
## 10 58.92484 2.604913 0.9009968
## 5  53.06525 2.466751 0.9446590
## 8  49.60126 2.484616 0.9089688
## 7  48.78769 2.428508 0.9164136
## 3  45.00871 2.328049 0.9090326
## 2  43.35530 2.344530 0.9277495
```

# likert Plots



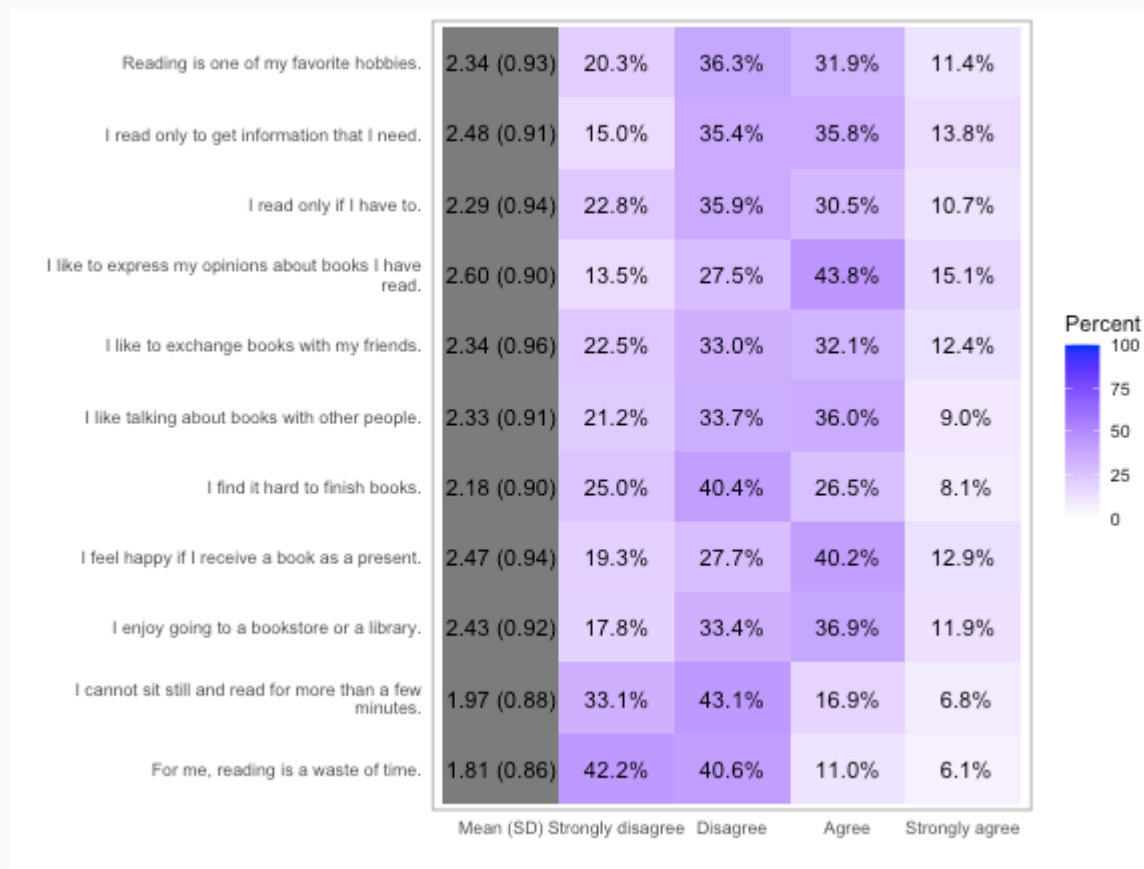
```
plot(l24)
```



# likert Plots



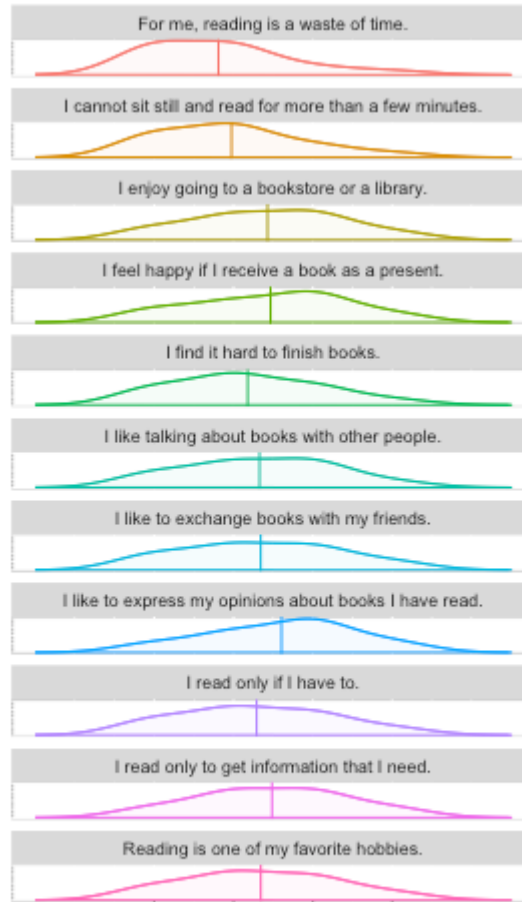
```
plot(l24, type='heat')
```



# Likert Plots



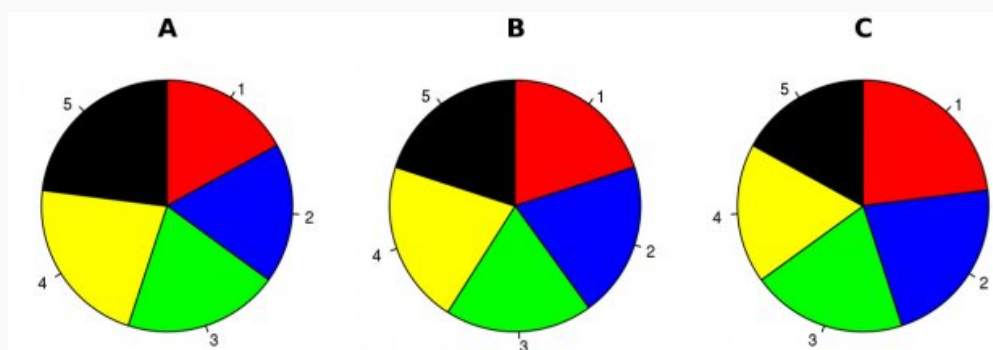
```
plot(l24, type='density')
```



File failed to load: <http://mathjax.rstudio.com/latest/jax/output/CommonHTML/jax.js>

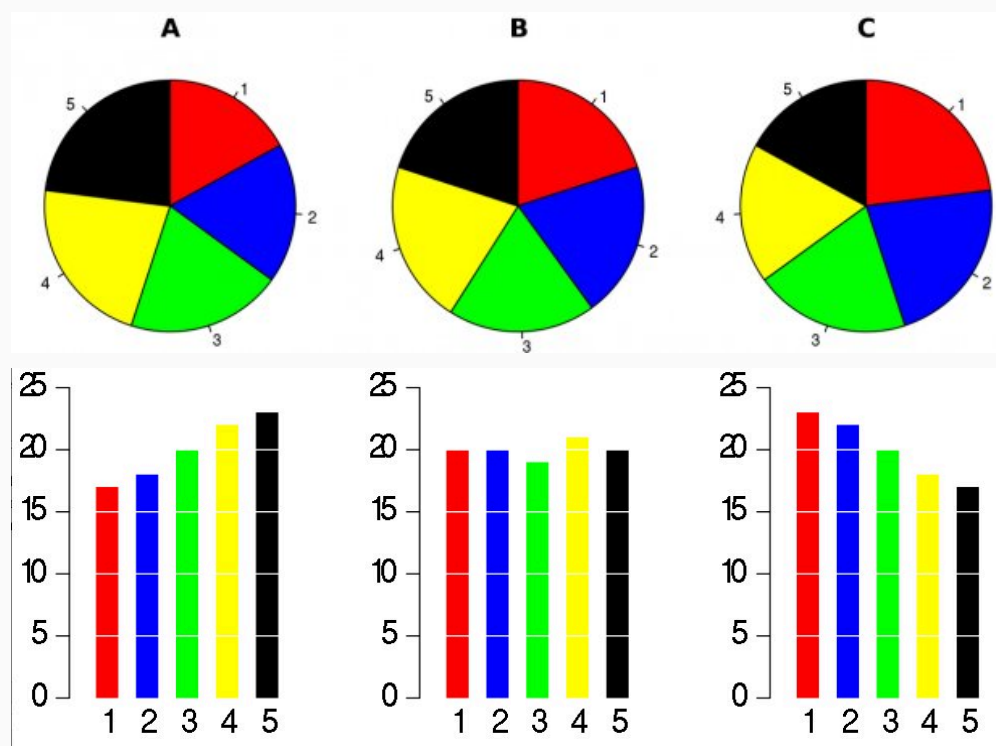
# Pie Charts

There is only one pie chart in *OpenIntro Statistics* (Diez, Barr, & Çetinkaya-Rundel, 2015, p. 48). Consider the following three pie charts that represent the preference of five different colors. Is there a difference between the three pie charts? This is probably a difficult to answer.



# Pie Charts

There is only one pie chart in *OpenIntro Statistics* (Diez, Barr, & Çetinkaya-Rundel, 2015, p. 48). Consider the following three pie charts that represent the preference of five different colors. Is there a difference between the three pie charts? This is probably a difficult to answer.



"There is no data that can be displayed in a pie chart that cannot better be displayed in some other type of chart"

John Tukey

# Additional Resources

For data visualization:

- ggplot2 website: <https://ggplot2.tidyverse.org>
- R for Data Science book: <https://r4ds.had.co.nz/data-visualisation.html>
- R Graphics Cookbook: <https://r-graphics.org>
- Data visualization cheat sheet: <https://github.com/rstudio/cheatsheets/raw/master/data-visualization-2.1.pdf>

# One Minute Paper

1. What was the most important thing you learned during this class?
2. What important question remains unanswered for you?



<https://forms.gle/bEk6KegZ1AMvQHnt5>